Karthick K

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EDUCATION

B.E. Computer Science and Engineering, Government College of Technology, Coimbatore — Expected 2029

2nd Year | CGPA: 9.32

HSC (12th), Oasis Matriculation Higher Secondary School — 2024 - 2025

96.6%

SKILLS

Programming: C, Python, Java (Basics)

Soft skills: Communication, Problem Solving, Team Coordination & Leadership

PROJECTS

Bus Scam Shield: Scam Website Detector (Chrome Extension) — Oct 2026

NextGen AI Hacks 2026, 24-hour hackathon, Rajalakshmi Engineering College | Team Leader (team of 4)

- Built a Chrome extension that checks the travel site is user is visiting and shows a green, yellow or red banner before they pay.
- Scores each site on 3 checks (domain age, SSL, look-alike domains, payment gateway, operator and GST/PAN verification) combined with a Random Forest model, Claude LLM analysis and Google Safe Browsing.
- Moved all API keys and checks to a Flask backend after jury feedback, so no key is exposed in the extension. Also added RSA-signed QR tickets and a scalping-risk check.

PlantCare: Crop Disease Detection App — Smart India Hackathon 2026

Problem Statement SIH2 6131: Early detection and management of crop diseases and pest infestations (Agriculture, FoodTech & Rural Development)

- Designed a mobile solution where a farmer photographs a leaf and gets the disease name and remedy in seconds, in their own regional language through voice output.
- Trained a MobileNetV2 transfer-learning model on the PlantVillage dataset (54,000+ images) that reached 93% accuracy across 38 crop disease classes.
- Fixed a 36x class imbalance with class-weighted loss instead of duplicating images, and used rotate, zoom and flip augmentation to match real farm photos.
- Added confidence-aware results: below 60% confidence the app asks for a clearer photo instead of giving an unreliable diagnosis.
- Converted the model to TensorFlow Lite s s detection runs offline on the phone, for farmers with little or no internet.

Rice Type Classification — Seminar project

- Predicted rice type from 10 grain-shape features (118.185 grains) by comparing 6 classifiers; SVM performed best at 99.04% cross-validated accuracy.

CERTIFICATIONS

- Python for Data Science, NPTEL (IIT Madras), Elite certificate, score 84% (Jul - Aug 2026)
- FLY-Scholar (Finding the Leader in You), Competitiveness Mindset Institute, USA (Aug 2026)

CLUBS AND ACTIVITIES

- Member, gSARC (Team ETA), Government College of Technology
- Volunteer, National Service Scheme (NSS), Government College of Technology